

Ali Kassem | Computer Science Student & AI Automation Engineer

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Summary

Driven Computer Science student at E-JUST specializing in AI Systems Architecture, Automation, and Software Engineering. Experienced in designing production-oriented, fault-tolerant multimodal AI platforms and intelligent workflow automation systems. Proficient in Python, n8n orchestration, LLM integration, and microservices architecture, with a strong focus on building high-availability technology solutions to solve real-world problems.

Education

E-JUST University

Bachelor of Computer Science

Specializing in Artificial Intelligence and Software Engineering.

Alexandria, Egypt

2024–2028

Victoria College

High School Diploma

Alexandria, Egypt

2010–2024

Experience

Digital Egypt Pioneers Initiative (DEPI)

Data Engineering Trainee

Hybrid, Egypt

Jul 2026–Present

- Selected for the DEPI specialization track in AI & Data Science – Microsoft Data Engineer, developing scalable cloud architectures and modern data solutions.
- Architected and implemented cloud data workflows and big data processing pipelines utilizing Microsoft Azure technologies.
- Designed end-to-end ETL/ELT data pipelines, relational database schemas, and data management systems using MySQL and Python.
- Integrated AI-driven data workflows, automated deployment strategies, and prompt engineering practices for production-grade data systems.

Xare AI Platform

Lead Developer & Systems Architect

Personal Project

2026–Present

- Designed and engineered Xare AI from scratch and expanded it into Xare AI V2, a production-oriented, zero-cost multimodal AI platform.
- Architected a 120-node n8n orchestration system featuring intelligent model routing and multi-tier fallbacks across Gemini, Qwen 3.6 27B, GPT-OSS 120B, Whisper, and Deepgram.
- Built specialized pipelines for advanced reasoning, vision processing, document/PDF OCR, real-time voice synthesis, and image generation.
- Developed React/TypeScript frontend with real-time WebSocket voice streaming, IndexedDB media state management, and cloud deployment infrastructure.

Supply Chain Management (SCM) Simulation

Lead Developer & Systems Architect

University Project

Jan 2025–Apr 2025

- Led the design and development of a comprehensive SCM simulation in Python using OOP principles, personally authoring 60% of the main codebase.
- Integrated a local LLM for automated business insights, customer review sentiment analysis, and marketing content generation.
- Architected complete desktop software solution featuring a Tkinter GUI, SQLite database persistence, and Proxy design patterns.

Youth Technology Academy (YTA)
Artificial Intelligence Intern
○ Contributed to the AI development lifecycle through data preparation, model training, and performance evaluation and testing.

Remote, Egypt
Dec 2024–Jan 2025

Certificate

Oracle <i>AI for You: Training and Assessment</i> Oracle AI for You: Training & Assessment Completion	Online <i>Aug 2026</i>
Microsoft <i>Introduction to Machine Learning Concepts</i>	Online <i>Dec 2025</i>
Microsoft <i>Introduction to AI Concepts</i>	Online <i>Dec 2025</i>
Huawei <i>HCIA-AI V4.0 Course</i> Credential ID: ICT20251126005697	Online <i>Nov 2025</i>
American Institute <i>English Conversation Advanced 3</i> Credential ID: 102855	Alexandria, Egypt <i>Jan 2023</i>

Skills

Programming: Python (Object-Oriented Programming), SQL, MySQL, TypeScript, JavaScript

Data & Cloud: Data Engineering, ETL/ELT Pipelines, Microsoft Azure, Big Data Processing, Relational Databases (MySQL, SQLite)

AI & Machine Learning: LLM Integration, Prompt Engineering, Multi-Modal Routing, Machine Learning, Image Classification, OCR

Orchestration & DevOps: n8n Workflow Automation, Systems Architecture, Microservices, Web-Sockets, DevOps, Fault Tolerance

Software & Tools: Hugging Face, Firebase, GitHub, React, Tkinter, IndexedDB

Languages: Arabic (Native), English (Advanced), Japanese (Basics)